INTERNSHIP REPORT

Time Series Analysis of Stock Prices Using Python

Internship Task

Level 2 (Intermediate)

Task 2: Time Series Analysis

1. Introduction

Time Series Analysis is a statistical technique used to analyze observations collected over time. It enables analysts to identify patterns such as trends, seasonality, and random fluctuations that can assist in forecasting and decision-making.

This internship task focused on analyzing stock market data using Python and statistical time series techniques.

2. Objectives

The objectives of this project were to:

- Analyze stock prices over time.
- Visualize historical stock price movements.
- Identify trends and recurring patterns.
- Decompose the time series into trend, seasonal, and residual components.
- Apply moving average smoothing to reduce noise.

3. Tools and Technologies Used

The following tools were used:

- Python
- Pandas
- NumPy
- Matplotlib
- Statsmodels
- Jupyter Notebook.

4. Dataset Description

The Stock Prices Dataset contains historical trading information for publicly traded companies.

Attributes included:

- Symbol
- Date
- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

For this analysis, the Close Price was selected because it best represents the final market valuation of the stock on a trading day.

5. Data Preprocessing

Loading the Dataset

The dataset was imported into a Pandas DataFrame.

Missing Value Analysis

A missing value assessment was conducted using:

df.isnull().sum()

The analysis revealed:

- Open Price: 11 missing values
- High Price: 8 missing values
- Low Price: 8 missing values

Missing values were found in the Open, High and Low Price column.

Missing Value Treatment

Forward-fill imputation was used:

df = df.ffill()

This method replaced missing values with the most recent valid observations.

Date Conversion

The Date column was converted into datetime format:

```
df['date'] = pd.to_datetime(df['date'])
```

Data Filtering

A single stock (AAPL) was selected for analysis.

6. Methodology

Step 1: Time Series Visualization

The Close Price was plotted against Date to visualize stock movements over time.

Step 2: Trend Identification

Visual inspection of the plot was used to identify overall market direction.

Step 3: Seasonal Decomposition

The seasonal_decompose() function from Statsmodels was applied to separate the series into:

- Trend Component
- Seasonal Component
- Residual Component

Step 4: Moving Average Smoothing

A 30-day moving average was calculated using:

```
rolling(window=30).mean()
```

This helped smooth short-term fluctuations and reveal the underlying trend.

7. Results and Discussion

Time Series Plot

The stock price displayed continuous fluctuations over time, reflecting normal market behavior.

Trend Analysis

The trend component showed the long-term movement of stock prices and indicated the general market direction.

Seasonal Analysis

Seasonal decomposition revealed recurring cyclical patterns within the dataset.

7.4 Residual Analysis

Residual values represented irregular market fluctuations not explained by trend or seasonality.

7.5 Moving Average Analysis

The 30-day moving average smoothed daily volatility and highlighted broader price movements.

8. Challenges Encountered

Several challenges were encountered:

- Installation of the Statsmodels library.
- Handling missing values.
- Selecting an appropriate decomposition period.
- Understanding the interpretation of trend and seasonal components.

These challenges were successfully resolved through data preprocessing and library installation.

9. Conclusion

This project successfully demonstrated the application of Time Series Analysis techniques to stock market data. The analysis revealed trends, seasonal behavior, and irregular fluctuations while showcasing the usefulness of moving averages for smoothing noisy financial data.

The task strengthened practical skills in data preprocessing, visualization, and time series modeling using Python.

10. Recommendations

Future studies can improve the analysis by:

- Comparing multiple stocks.
- Applying forecasting techniques such as ARIMA.
- Incorporating external economic indicators.
- Using advanced machine learning models for prediction.